*MA.8.AR.2.3*

## Benchmark

| | |
|---|---|
| **MA.8.AR.2.3** | **Given an equation in the form of $x^2 = p$ and $x^3 = q$, where $p$ is a whole number and $q$ is an integer, determine the real solutions.** |

Benchmark Clarifications:
*Clarification 1:* Instruction focuses on understanding that when solving $x^2 = p$, there is both a positive and negative solution.
*Clarification 2:* Within this benchmark, the expectation is to calculate square roots of perfect squares up to 225 and cube roots of perfect cubes from -125 to 125.

## Connecting Benchmarks/Horizontal Alignment

- MA.8.NSO.1.1, MA.8.NSO.1.7
- MA.8.GR.1.1, MA.8.GR.1.2

## Terms from the K-12 Glossary

- Integer
- Real Numbers
- Whole Numbers

## Vertical Alignment

**Previous Benchmarks**
- MA.7.AR.2.2

**Next Benchmarks**
- MA.912.AR.3.1

## Purpose and Instructional Strategies

In grade 7, students wrote and solved two-step equations in one variable. In grade 8, when given an equation in the form $x^2 = p$ and $x^3 = q$, where $p$ is a whole number and $q$ is an integer, students determine the real solutions. In Algebra 1, students will write and solve quadratic equations.

- This benchmark involves students understanding the concepts of how to square a number and find the square root as well as how to cube a number and find the cube root.
- Students should recognize that squaring a number and taking the square root of a number are inverse operations, therefore, cubing a number and taking the cube root are inverse operations as well. Students should use this understanding to solve equations containing square or cube numbers.
- In finding the square root, instruction involves discussion that there is both a positive and negative solution. Instruction can include relating the lengths of the sides of a square for square root and the length of the side of a cube to cube roots.
- Within this benchmark, it is not the expectation that students are required to isolate the $x^2$ term or the $x^3$ term when solving an equation.

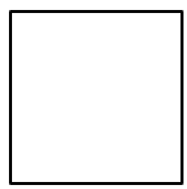
## Common Misconceptions or Errors

- Students may incorrectly conclude that squaring a number means to multiply by 2. Likewise, cubing may be mistaken as multiplying by 3. Use length to show doubling and area of a square to show an exponent of 2. Use of two-dimensional and three-

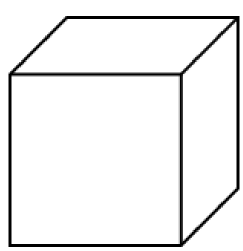
dimensional manipulatives (*MTR.2.1*) may also help to emphasize squares and cubes versus increasing length.
- Students may think that since a negative number has no square root in the real number system, then a negative number has no cube root in the real number system.

## Strategies to Support Tiered Instruction

- Instruction includes modeling the differences between doubling and squaring a value using a graphic organizer. Doubling a value would be represented by multiplying a given length by 2 whereas squaring a number would be represented by the area of a square with a given length.
  - For example, students can be given the table below to show how the left column doubles a length whereas the right column squares a length.

| Doubling | Squaring |
|---|---|
| Given length of 5<br><br>Representing $5 \cdot 2$ | Given length of 5<br><br>Representing $5^2$<br> |

- Instruction includes modeling the differences between tripling or cubing a value using a graphic organizer. Tripling a value would be represented by multiplying a given length by 3, whereas cubing a number would be represented by the volume of a cube with a given length.
  - For example, students can be given the table below to show how the left column triples a length whereas the right column cubes a length.

| Tripling | Cubing |
|---|---|
| Given length of 5<br><br>Representing $5 \cdot 3$ | Given length of 5<br><br>Representing $5^3$<br> |

- Instruction may include providing students with the opportunity to develop their own note sheet or graphic organizer for the cubes of numbers from -5 to 5.

| $x$ | $x^2$ | $x^3$ |
|---|---|---|
| $-5$ | $25$ | $-125$ |
| $-4$ | $16$ | $-64$ |
| $-3$ | $9$ | $-27$ |
| $-2$ | $4$ | $-8$ |
| $-1$ | $1$ | $-1$ |
| $0$ | $0$ | $0$ |
| $1$ | $1$ | $1$ |
| $2$ | $4$ | $8$ |
| $3$ | $9$ | $27$ |
| $4$ | $16$ | $64$ |
| $5$ | $25$ | $125$ |

## Instructional Tasks

*Instructional Task 1(MTR.7.1)*

A square tile in a kitchen has an area of 121 square inches.

Part A. What is the length of one side of the square tile in inches? Is this tile smaller or larger than a one-foot by one-foot tile?

Part B. The owner of the house, Kiana, wants to put larger tiles in their kitchen to change the look of the kitchen. The new tile is a square with an area of 196 square inches.

- What is the length of the side of the new tile?
- How does this larger tile compare to the current tile used in the kitchen?

Part C. A third tile has a side length of $2\sqrt{11}$. Kiana is trying to determine which square tile covers the most area. Put the tile's side lengths of the tiles in order from greatest to least. Justify your thinking.

*Instructional Task 2 (MTR.3.1)*

The volume of a large cube is 125 cubic inches. The volume of a small cube is 27 cubic inches. What is the difference between the length of one side of the large cube and the length of one side of the small cube?

## Instructional Items

*Instructional Item 1*

An equation is given.

$$x^2 = 49$$

What are the values of $x$?

*Instructional Item 2*

Solve for $b$ in the equation $-64 = b^3$.

**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*